Bank Database — ACID Properties & Transaction Isolation

SQL PRACTICAL / LAB EXERCISE

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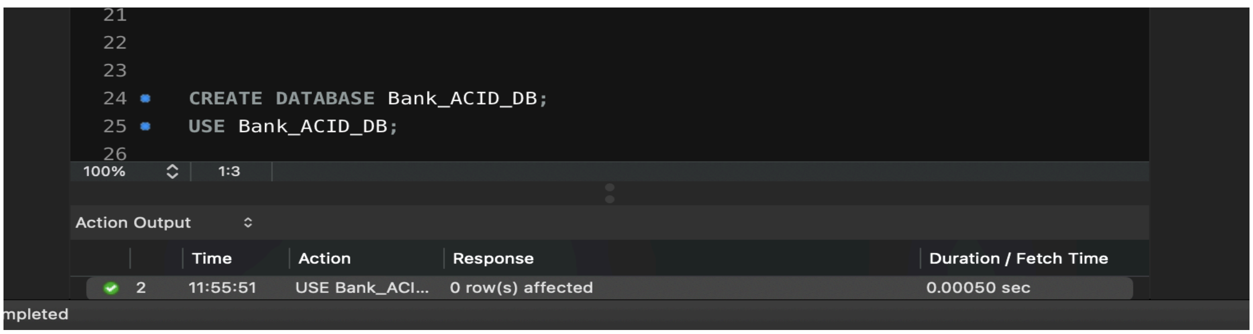
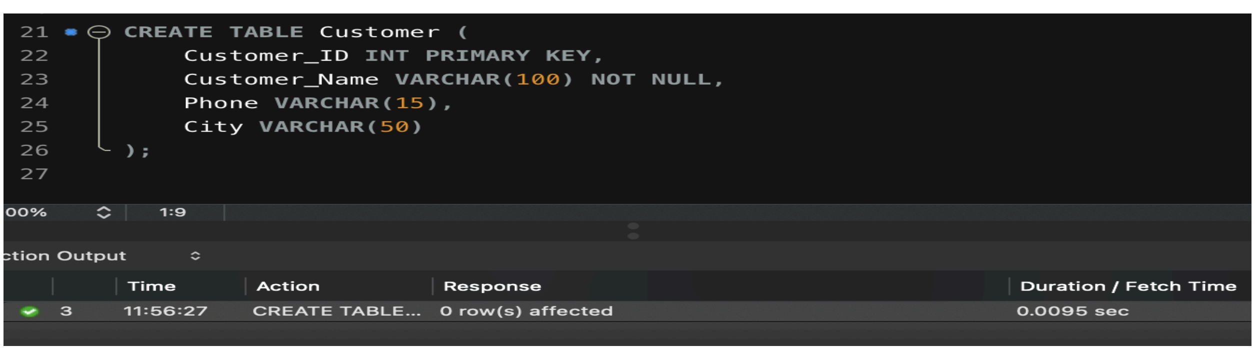
Step 1 — Create Database

```
CREATE DATABASE Bank_ACID_DB;
USE Bank_ACID_DB;
```



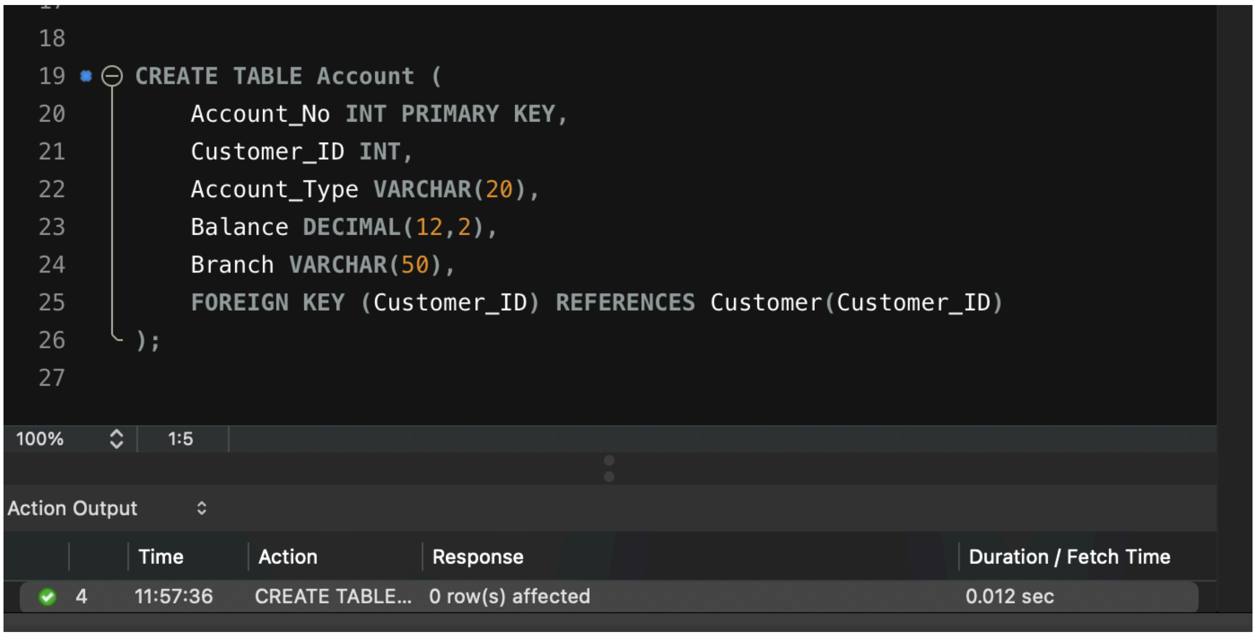
Step 2 — Create Customer Table

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    City VARCHAR(50)
);
```



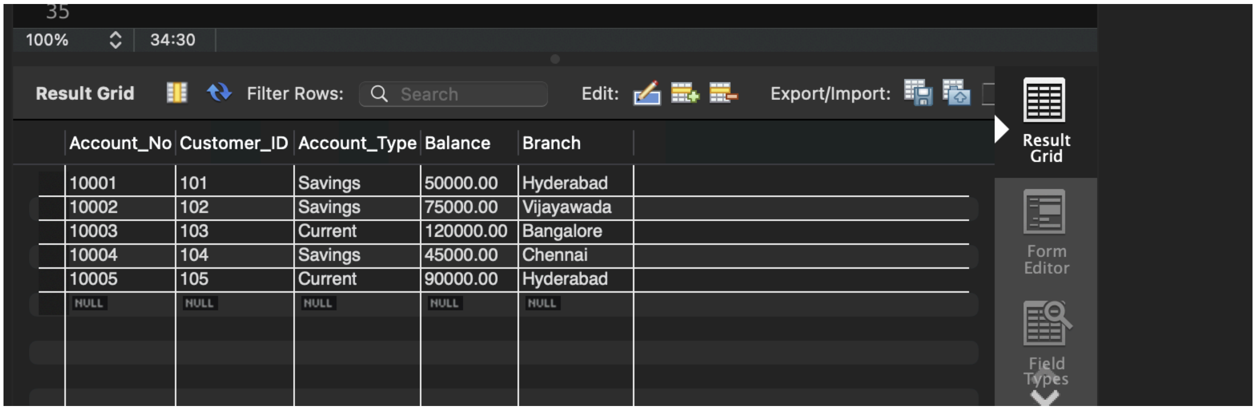
\Step 3 — Create Account Table (FOREIGN KEY)

```
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```



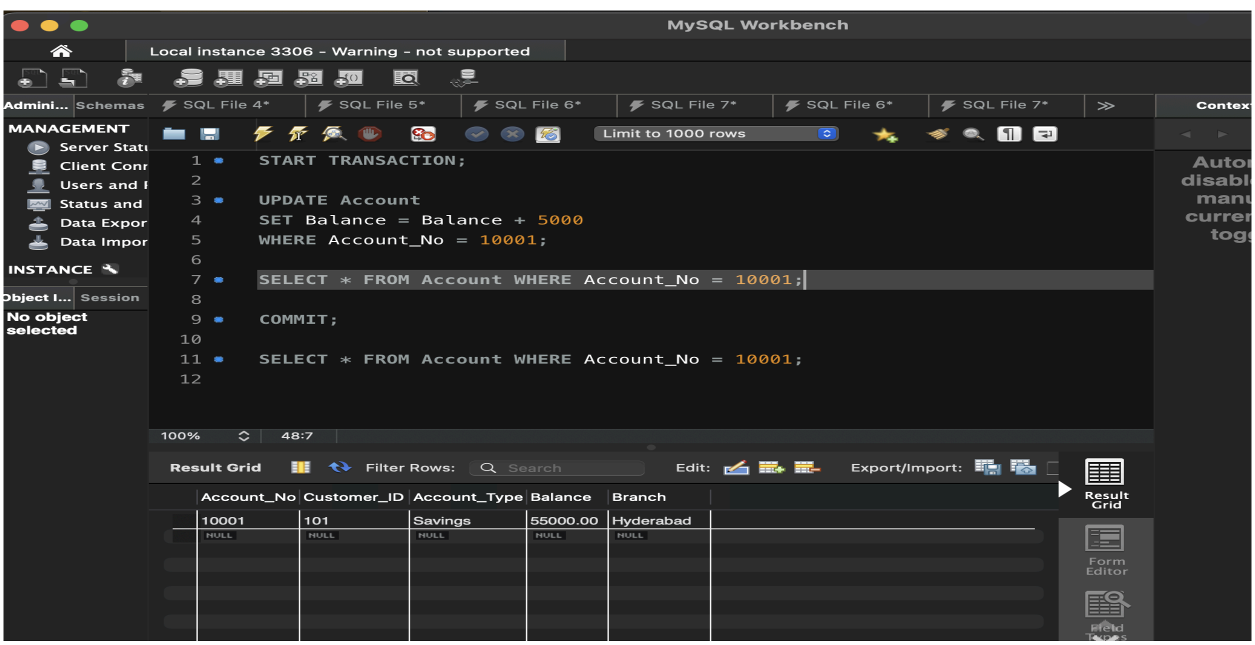
Step 4 — Insert Customer & Account Data, Then Display

```
INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES
(101, 'Hansika Reddy', '9876543210', 'Hyderabad'),
(102, 'Priya Sharma', '9876543211', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),
(104, 'Sneha Rao', '9876543213', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');

INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');

SELECT * FROM Customer;
SELECT * FROM Account;
```



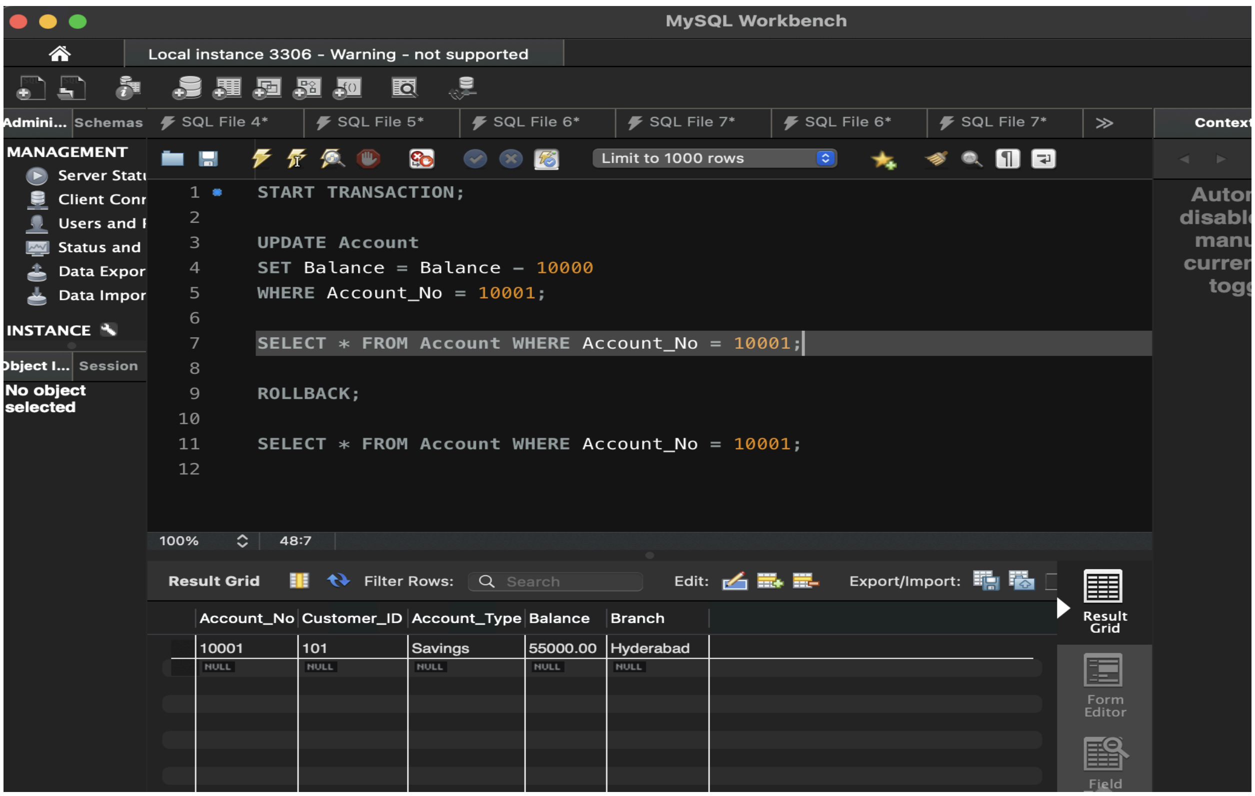
Step 5 — Deposit Transaction (COMMIT)

```
START TRANSACTION;

UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;

SELECT * FROM Account WHERE Account_No = 10001;

COMMIT;

SELECT * FROM Account WHERE Account_No = 10001;
```



Step 6 — Withdraw Transaction (ROLLBACK)

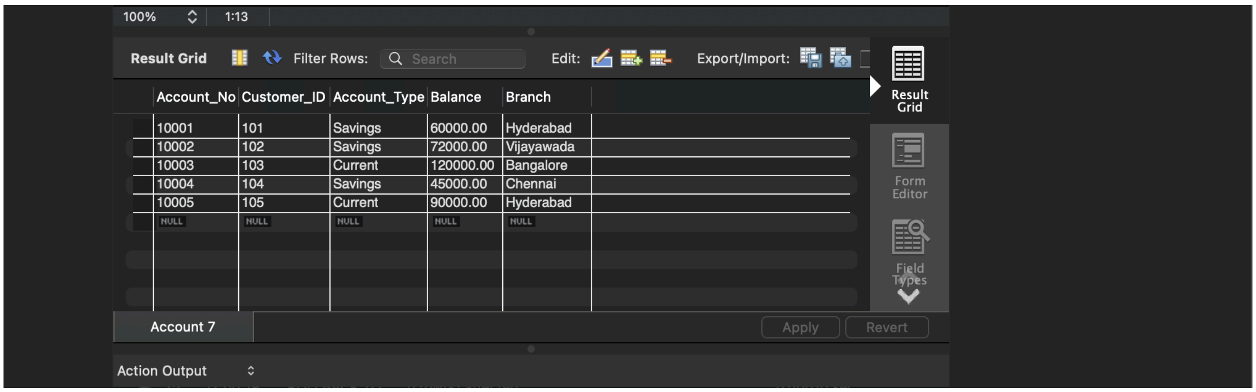
```
START TRANSACTION;

UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;

SELECT * FROM Account WHERE Account_No = 10001;

ROLLBACK;

SELECT * FROM Account WHERE Account_No = 10001;
```



Step 7 — SAVEPOINT Demonstration

```
START TRANSACTION;

UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001;

SAVEPOINT Deposit1;

UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002;

SAVEPOINT Withdrawal1;
```

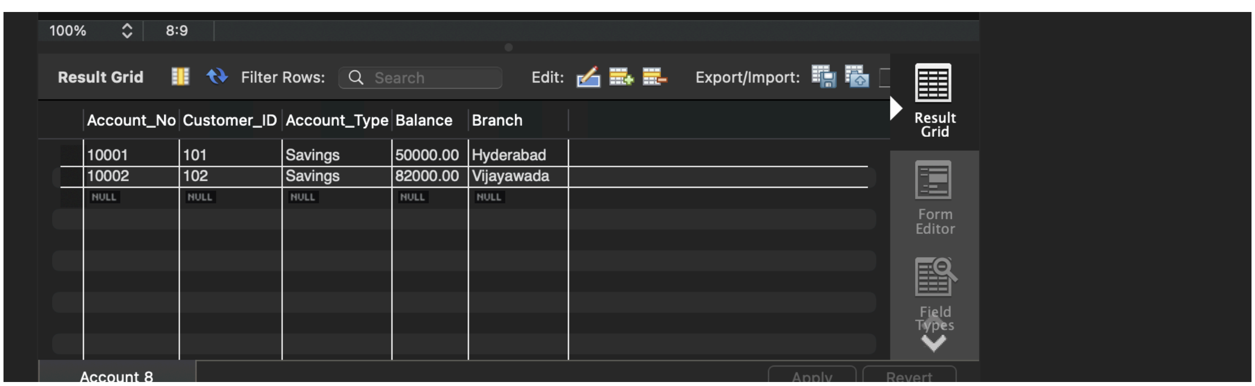
```
UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003;

ROLLBACK TO SAVEPOINT Withdrawal1;

COMMIT;

SELECT * FROM Account;
```



Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	72000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Step 8 — Bank Transfer (Demonstrates ATOMICITY)

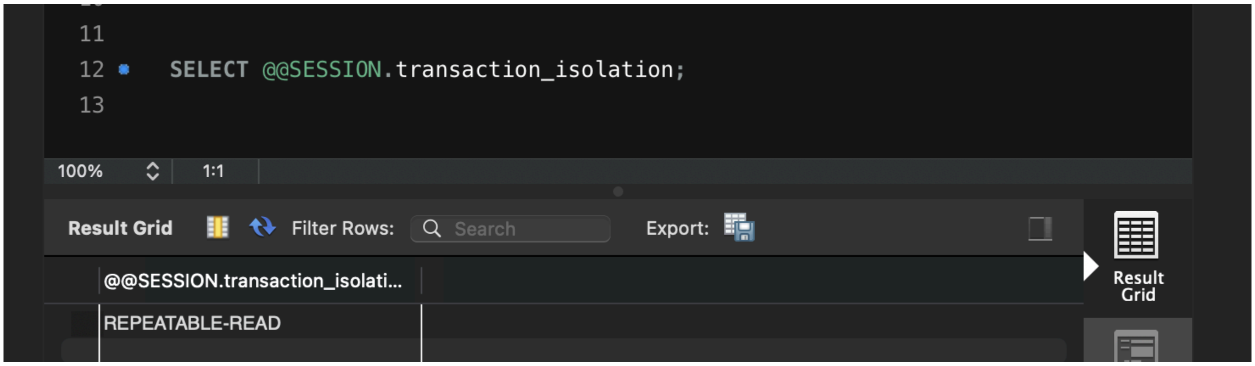
```
START TRANSACTION;

UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001;

UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002;

SELECT * FROM Account WHERE Account_No IN (10001,10002);

COMMIT;
```



Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

Step 9 — Check Current Transaction Isolation Level

```
SELECT @@SESSION.transaction_isolation;
```



Step 10 — Change Isolation Level to READ COMMITTED

```
SET SESSION TRANSACTION ISOLATION LEVEL READ COMMITTED;

START TRANSACTION;

SELECT * FROM Account WHERE Account_No = 10001;

COMMIT;
```